

TECHIN 510 — Week 1 Lab Report

Week 1 Lab — Submission Report (Components A–E + Reflection)

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Course / Section: TECHIN 510
GitHub repo (Classroom): <https://github.com/GIX-Luyao/lab-5-c4han-star>
Personal mirror (optional): <https://github.com/c4han-star/gix-week1-lab>
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Component A — Deliverables

Interview notes + synthesis: see COMPONENT_A.md in this repository (workflow, BlueTally, pain points, improvement ideas, and **problem statement**).

Problem statement (lab template):

When GIX staff running the post-launch asset return desk needs to reconcile checked-out equipment and newly purchased items against BlueTally, they currently verify and enter assets item-by-item at return time and often apply UW barcodes only after intake, which causes slow queues, duplicate handling, and errors such as missing pieces or unclear Makerspace-vs-IT routing.

Component B — Deliverables

1. AI usage log (minimum 3 interactions)

#	Prompt (summary or exact)	What the AI produced	First try?	Fixes / follow-ups
1	Build Streamlit app from Component A themes: asset return / BlueTally FAQ guide, 10–15 entries,	app.py, requirements.txt, .gitignore, .streamlit/config.toml	Mostly yes	Tuned categories (BlueTally vs routing vs kits); ensured exactly

#	Prompt (summary or exact)	What the AI produced	First try?	Fixes / follow-ups
	categories, search, Home/About; align copy with interview notes.			one <code>st.title()</code> per view.
2	Add sidebar radio Home/About; About describes prototype scope (no BlueTally API); link to GIX site; semantic headings.	Updated <code>app.py</code> navigation and About copy	Yes	Minor wording edits only.
3	Wayfinder: dict-based resources, sidebar category filter, search, empty state, <code>assert</code> on required keys for each resource.	<code>wayfinder_app.py</code>	Yes	Confirmed <code>assert</code> runs at import; clarified captions for hints.

2. Accessibility baseline

Color contrast

- **Check:** Body text vs page background using Streamlit default light theme; primary widgets use theme primary #1f77b4 from `.streamlit/config.toml`.
- **Tool:** [WebAIM Contrast Checker](#) — sampled #262730 on #ffffff (body) and primary button styling via browser inspector.
- **Result: Pass** for normal body text (approximate ratio meets WCAG AA for typical Streamlit defaults); custom primary is a medium blue on white—spot-check passed for large controls.
- **Fix:** None required.

Semantic headings

- **Check:**
- [x] One `st.title()` per view (**Home** shows “GIX Asset Return & BlueTally Field Guide”; **About** shows “About this project”).
- [x] Sections use `st.header()`; expanders use captions + `st.write`, not fake markdown heading lines as titles.
- [x] Order: title → header (“Search”, “Matching questions”) → content.
- **Result: Pass** — headings match Streamlit’s mapped roles (h1/h2).

Component C — System architecture & design

1. I/P/O diagram — Component B app (`app.py`)

flowchart LR

subgraph Input

I1[Search text]

I2[Category selection]

I3[Page choice Home/About]

```

end
subgraph Process
  P1[Filter FAQ list by substring + category]
  P2[Render expanders / About markdown]
end
subgraph Output
  O1[Filtered FAQ entries]
  O2[About copy + external link]
end
I1 --> P1
I2 --> P1
I3 --> P2
P1 --> O1
P2 --> O2
O1 -->|User refines query or changes category| I1

```

2. Design decision log

Field	Your entry
Decision	Single-file Streamlit app (app.py) for the Asset Return / BlueTally Field Guide.
Alternatives considered	Split FAQs into faq_data.py; add pages/ multipage API; merge Wayfinder into same file.
Why you chose this	Keeps Week 1 scope small; easy for graders to open one file; matches how the lab introduces Streamlit.
Trade-off	Data + UI in one place will get crowded past ~300–400 lines or multiple collaborators.
When choose differently?	Shipping multiple unrelated flows or adding authentication—then modular layout (pages/) or packages.

Component D — Testing & validation

Smoke test table

Environment: macOS, Python 3 with .venv, Streamlit from requirements.txt.
Verified 2026-05-04.

#	Feature tested	Action you took	Expected result	Actual result	Pass/Fail
1	Search	Typed barcode in “Find an answer”	FAQs mentioning barcode / intake timing	Four matching expanders (hits across BlueTally + Improvement Ideas, etc.)	Pass

#	Feature tested	Action you took	Expected result	Actual result	Pass/Fail
2	Category filter	Sidebar category BlueTally only	Only rows where category == BlueTally	Three entries (“What is BlueTally...”, Asset vs Accessory, When do barcodes...) Page explains prototype scope; markdown link to https://gixnetwork.org/ renders	Pass
3	About page	Sidebar About	About text + GIX link		Pass

Screenshot

See repository folder **screenshots/** — `component-b-home.png` captures the Home view with sidebar navigation (smoke-test evidence).

Quality gate checklist

- ☒ Smoke test table completed (3 features)
- ☒ No failing tests remaining (all Pass)
- ☒ Screenshot included (`screenshots/component-b-home.png`)
- ☒ Accessibility baseline recorded (contrast + headings)

Component E — GIX Wayfinder

1. I/P/O diagram (Wayfinder)

```

flowchart LR
    subgraph Input
        A[Search string]
        B[Category select]
    end
    subgraph Process
        C[Load RESOURCES list]
        D[Filter by substring + category]
    end
    subgraph Output
        E[Expanders with details + hints]
        F[No-results warning]
    end
    A --> D
    B --> D
    C --> D
    D --> E
    D --> F
    E -->|User edits query or category| A
  
```

2. Edge cases (document 2)

Edge case	Why it matters	What you tested	Result
Empty search + All categories	Confirms full dataset loads and UI renders every resource	Blank search, category All	Seven expanders listed (makerspace, bike storage, printing, etc.) — Pass
Nonsense query zzz123	Empty-state messaging; no silent empty page	Typed zzz123	Yellow warning: “No resources match...” — Pass

3. Assert statement (data integrity)

- **File:** `wayfinder_app.py`
- **Function:** `assert_resources_integrity` — called immediately after `RESOURCES` is defined.
- **Checks:** Each dict has non-empty name and category strings.

4. Prompt log excerpt

- **Initial prompt:** “Streamlit campus Wayfinder: list of dict resources with name, category, area, details, hint; sidebar category filter; search box; expanders; warn when no matches; add assert that every resource has name + category.”
- **Refinement:** “Use `st.caption` for wayfinding hint; keep expanders expanded for readability on first paint.”
- **Why:** Improves scanability and matches lab expectation of clear empty-state behavior.

Reflection (Component B)

1. What surprised you about AI-assisted coding?

The speed of a first runnable Streamlit skeleton was surprising—within one interaction you get files on disk, not just chat text. The “wow” moment was seeing filters and navigation wired without writing boilerplate by hand. The “wait” moment was remembering to verify semantics (heading hierarchy, real-world accuracy of FAQ copy) because the model can sound authoritative while glossing edge cases.

2. What did the AI get wrong?

Early drafts sometimes mixed generic “purchasing” language that did not match the interview. Fixing it meant pasting the interview summary back in and tightening prompts. No runtime crashes after pinning `streamlit>=1.30` and keeping imports minimal.

3. Could you explain your code?

filter_entries in `app.py` lowercases the query and each FAQ's question/answer/category into a single string, then keeps rows where the query is empty or appears as a substring. Category filter is applied first so the search only runs on the selected slice—same idea as `filter_resources` in the Wayfinder.

4. What did you learn from the interview?

The return desk is not just “inventory”—it is **two mental workflows** (checkout vs newly purchased) stacked on the same night, plus **late barcoding**. That pushed the FAQ toward reconciliation language, routing ambiguity, and kit checklists instead of a generic order form.

Honor statement

I followed the course collaboration policy. This submission is my own work; AI tools were used as permitted for implementation and editing.

— Finnick Chen — 2026-05-04